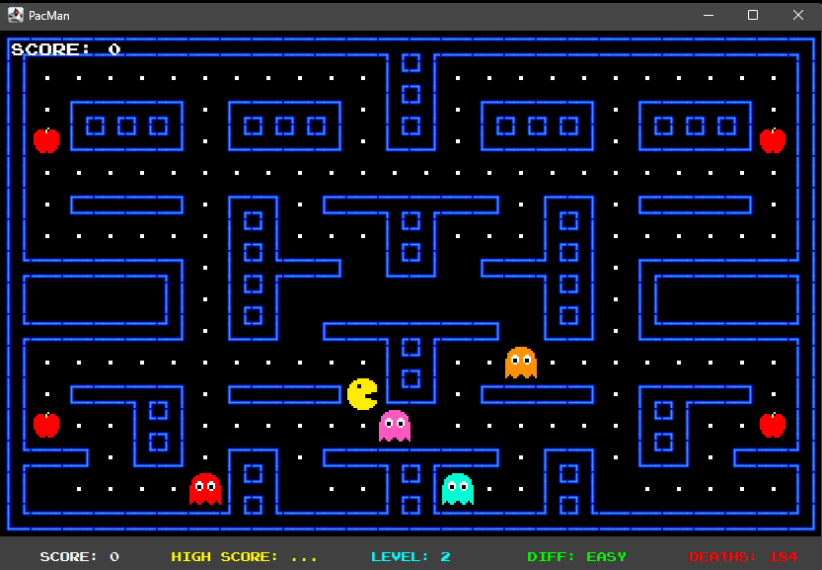


PACMAN OOP PROJECT REPORT

STUDENT: Roghaye Saadabadi

STUDENT ID: 404463707

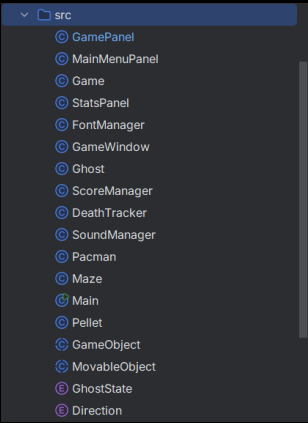
OVERVIEW



PROJECT REQUIREMENTS

Classes

- Game ✓
- Pacman ✓
- Ghost ✓
- Maze ✓
- Pellet ✓
- ScoreManager ✓



Environment

Game Start - Game Exit



We have a main menu for both Starting the Game, Exiting the game, Selecting Game Mode and Selecting Level which will be discussed in later sections

Levels are defined in level folder using Ascii Characters

```
1 #####
1 #.....##.....#
2 #.####.####.####.####.#
3 #O####.####.####.####O#
4 #.....#.....#
5 #.####.####.####.####.#
6 #.....##...##...##.....#
7 #####.####.#.####.#####
8 #  #.##      ##.#  #
9 #####.##.#####.##.#####
10 #.....#.#.....#
11 #.####.#####M####.####.#
12 #O..##.....#.....##..O#
13 ###.##.##.#####.##.##.###
14 #B....##P..##I..##C....#
15 #####
```

= wall
M = Pacman Spawn Point
. = Normal Bean
O = Super Pellet (printed as apple fruit)

P = Pinky Ghost
I = Cyan Blue Ghost
C -> Orange Ghost
B -> Red Ghost

Walls, Beans, Pacman, Ghost, HighScore



- Walls, Beans, Pacman, Ghosts are all defined in map Ascii representation
- HighScore is saved in `highscore.txt` file

```

1 #####
2 #.....#
3 #O###.###.###.###O#
4 #.....#
5 #.###.###.###.###.###
6 #.....#
7 #####.###.###.###
8 #  ##    ##  #
9 #####.###.###.###
10 #.....#
11 #.###.###M###.###.###
12 #O.###.....###.O#
13 ###.###.###.###.###
14 #B.....#P..#I..#C.....#
15 #####

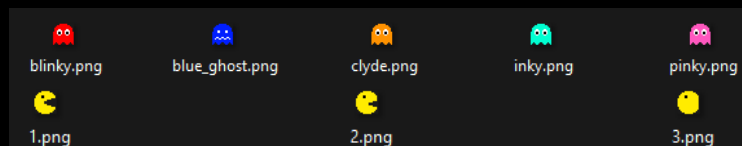
```

Maps

as discussed in previous section we can dynamically define maps in levels folder using Ascii Characters

Pacman Style

All assets can be found in Assets folder under main directory of project



Pacman Movement

- you can both control the Pacman using WASD keys or arrow keys
- Pacman can't bypass the walls
- only moving in allowed directions is allowed

Scoring System

- each key press would lead in one negative score

```
o if (maze.isValidMove(nextX, nextY)) {  
    this.x = nextX;  
    this.y = nextY;  
    this.currentDirection = nextDirection;  
  
    scoreManager.addScore(-1);  
}
```

-

- Collecting beans using Pacman would lead in bean deletion

```
o for (Pellet p : maze.getPellets()) {  
    if (p.isActive() && p.getX() == pacman.getX() && p.getY() ==  
        pacman.getY()) {  
        eatenPellet = p;  
        p.setActive(false);  
        scoreManager.addScore(10); // Requirement 4c  
    }  
}
```

- Showing Score during the game

SCORE: 63

- adding 500 scores on collecting all beans

```
o if (maze.getPellets().isEmpty()) {  
    System.out.println("YOU WIN! All beans collected.");  
    scoreManager.addScore(500); // Requirement 4e  
    scoreManager.checkAndSaveHighScore();  
    isGameOver = true;  
}
```

Ghost

- they can move wisely or randomly based on the difficulty level
(on Hard Mode they use A* algorithm and they are nearly unbeatable)
- they can't bypass walls
- they reroute wisely
- they can collide and kill Pacman

Ending

you should either collide with a normal ghost (not frightened) or collect all beans



You can also pause the game using escape key

Collision

- Pacman and ghosts can't bypass walls or the border defined in the map
- you would stop moving on collision with walls
- ghosts choose new direction on collision with walls
- beans get deleted on collision with Pacman
- beans would lead in score increasing during the game

Saving Score

we store highest score of the player in highscore.txt file in main project folder

Documentation

How to run the project

you can the project by easily cloning the repository and running mian.java.

Classes

they are fully described in comments of the source codes

Git

I have used git repeatedly during the project you can see meaningful commits during development

<https://github.com/artemiss199/Pacman>

```
commit 16a0973 (HEAD -> master, origin/master)
Author: artemiss199 <artemissdabadi@gmail.com>
Date: Tue Aug 11 19:22:59 2026 +0330

    FEAT[Levels]: we now read levels dynamically from levels folder instead of hardcoding it

commit ba600dc
Author: artemiss199 <artemissdabadi@gmail.com>
Date: Tue Aug 11 15:33:01 2026 +0330

    ADD[Game]: pause menu using ESC key

commit 857a8U3
Author: artemiss199 <artemissdabadi@gmail.com>
Date: Tue Aug 11 15:03:02 2026 +0330

    STYLE[Lose/Win]: change style of the text shown on screen on win and lose condition

commit b780779
Author: artemiss199 <artemissdabadi@gmail.com>
Date: Tue Aug 11 14:18:15 2026 +0330

    FEAT[Map]: ghosts and pacman are not hardcoded into the code anymore and can be represented in map

commit 25ff66e
Author: artemiss199 <artemissdabadi@gmail.com>
Date: Tue Aug 11 13:30:05 2026 +0330

    STYLE[MainMenu]: changed buttons and dropdown menus styles
    ... skipping ...
commit 16a0973 (HEAD -> master, origin/master)
Author: artemiss199 <artemissdabadi@gmail.com>
Date: Tue Aug 11 19:22:59 2026 +0330

    FEAT[Levels]: we now read levels dynamically from levels folder instead of hardcoding it

commit ba600dc
Author: artemiss199 <artemissdabadi@gmail.com>
Date: Tue Aug 11 15:33:01 2026 +0330

    ADD[Game]: pause menu using ESC key

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Author: artemiss199 <artemissdabadi@gmail.com>
Date: Tue Aug 11 15:03:02 2026 +0330

    STYLE[Lose/Win]: change style of the text shown on screen on win and lose condition

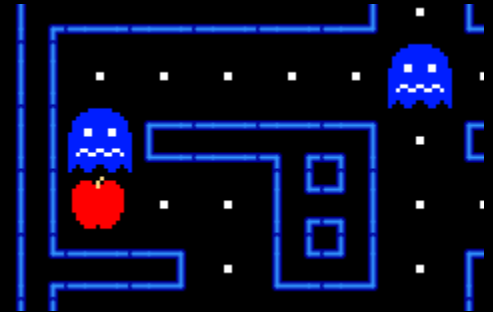
commit b780779
Author: artemiss199 <artemissdabadi@gmail.com>
```

Additional

Power Pellet

power pellets are represented as apples in the game you can eat them and make ghosts frightened

```
private void checkGhostCollisions() {
    // Pacman hits Ghost (Requirement 6a)
    for (Ghost g : ghosts) {
        if (g.getX() == pacman.getX() && g.getY() ==
pacman.getY()) {
            if (g.getState() == GhostState.FRIGHTENED) {
                soundManager.playSound("eatghost");
                scoreManager.addScore(200);
                g.respawn();
            } else {
```

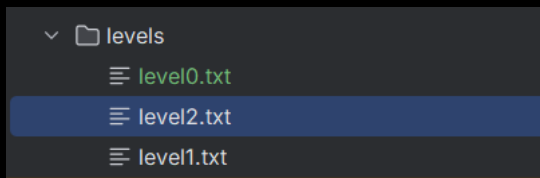


you can receive 200 scores by eating a frightened ghost

ghosts would jump back to their spawn point immediately after being eaten

Multi Level

you can define you own custom map in levels folder as I did



```
# = wall
M = Pacman Spawn Point
. = Normal Bean
O = Super Pellet (printed as apple fruit)
```

```
P = Pinky Ghost
I = Cyan Blue Ghost
C -> Orange Ghost
B -> Red Ghost (Blinky)
```

So, you can have as many ghosts as you want

```
1 #####
2 #.....##.....#
3 #.####.####.####.####.
4 #O####.####.####.####O#
5 #.....#
6 #.####.####.####.####.
7 #.....##...##...##.....#
8 #####.#### ## ####.#####
9 #   #.##      ##.##   #
10 #####.## ##### ##.#####
11 #.....#.....#
12 #.####.####M##.####.####.
13 #O..##.....  .....##..O#
14 ###.##.##.#####.##.##.###
15 #B.....#P..##I..##C.....#
16 #####
```

Music

all sound effects can be found in Assets/sfx folder

we have sfx for:

- Main menu
- Game start
- Pacman eating beans
- Pacman eating ghost
- Losing

Advanced Tracking Algorithms

we use A* with a bit of inaccuracy and mis-prediction (dumb acting) based on the difficulty level

if you want to use 100% accuracy, just put the difficulty on Hard in the main menu.

you can find fully explained implementation in Ghost.java file with explanation on the code

```
private void calculateAStarPath(Maze maze) { 1 usage  👤 artemiss199
    int targetX = target.getX();
    int targetY = target.getY();
```